



Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 & Discrete Time Signal Processing

Name of the Faculty member (s): Mrs.S.Jeeva

Innovative Practice Description

- Unit / Topic: III Finite Impulse Response Filters – Windowing Technique

- Course Outcome:

CO3 : The students will be able to Design Finite Impulse response (FIR) digital filters.

- Topic Learning Outcome:

TLO17 : Design FIR filters using windowing techniques.

TLO19: Realize the FIR filter design using direct form, linear phase and cascade realization.

- Activity Chosen: Team Assisted Individualization (TAI)

- Justification:

Team Assisted Individualization (TAI), was found to improve the academic behavior of the students, to increase their Problematic knowledge more than traditional methods. Three types of windowing problems are available for designing of FIR filter. More number of formulae are used for windowing method so students are must practice to solve effectively. Due to this activity the students are able to solve the problem easily.

- Time Allotted for the Activity: 2 periods (Each will carry 50 minutes)

- Details of the Implementation:

At first the students are divided into groups of four or 5 members (10 Groups). Each group name can be allotted by using Random Name Picker like spinning wheel (generated from classtools.net - https://www.classtools.net/random-name-picker/81_c4YLM8). Neat elaborated the TAI activity objectives and outcomes. Ground rules, rubrics are explained and provided the problem to be solved and materials.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PSO3
CO3	2	3	3	2	3	2	3	3	3

- PO / PSO mapped:

Innovative practice	PO4	PO5	PO8	PO9	PO10	PSO3
	2	3	2	3	3	3
Justification for correlation	The course outcome is moderately correlated since it	This course outcome is highly correlated since, the	Identify tenets of the professional code of ethics while doing	A collaborative activity TAI is planned which involves Implement the	The topics deals design of FIR filter involves Magnitude and phase plots to	The concept of Filtering operation is required for design, analyze

involves the performance analysis of filter magnitude and phase response, Matlab simulation tools is used to make measurements of filter characteristics	design of digital filter for the given specification is validated using MATLAB simulation tools and choose suitable technique.	activity-based learning. So, it is moderately correlated	norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work, to accomplish a goal. This course outcome is correlated highly.	complement writing and presentations. This task is accomplished through assignment therefore course outcome is correlated highly.	VLSI and communication systems. So, the course outcome is highly correlated.
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• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students feel that it was interesting. They learn how to communicate with team members and work together.

❖ **Benefit of the practice:**

The students' feel comfortable to learn in group. Make the students in teaching process more interesting. Each students got equal opportunity to come forward to take part in this activity. Increases students' learning achievement. The success of the activity was evaluated by asking the same model question in Internal Assessment test II – Around 75% of students answered correct.

❖ **Challenges faced in implementation:**

The main challenge faced is group conflict, must carry out supervision of each group in every learning implementation. Few Students are not ready to work

together. when the students face complicated problems in the small group, the teacher is still available to solve them.

References:

- <https://eric.ed.gov/?id=ED232852>
- <https://psycnet.apa.org/record/1985-10827-001>
- Slavin, R. E., Madden, N. A., & Leavey, M. (1984). Effects of team assisted individualization on the mathematics achievement of academically handicapped and nonhandicapped students. *Journal of Educational Psychology*, 76(5), 813–819. <https://doi.org/10.1037/0022-0663.76.5.813>
- John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHL. 2003.

Collaborative Learning

Team Assisted Individualization (TAI)

Course Code & Title: EC8553 Discrete Time Signal Processing

Degree, Semester & Branch: V Semester B.E. ECE A

Class Strength: 41 students

Group Name:

Register Number:

Topic chosen: Design of Finite Impulse Response filter using Windowing Method

Objectives of the Collaborative Activity:

The students will be able to

1. Design FIR filters using Windowing method.
2. Differentiate various windowing techniques.
3. Realize the filter using Direct form and Linear phase realization.
4. Simulate the designed filter using Matlab simulation tool for LPF, HPF, BPF and BSP.
5. Participate actively in the filter design learning process.
6. Discuss the concepts, convey ideas, share and verify the solutions, correct the mistakes and expertise to really understand the design procedure.
7. Eliminate the feelings of isolation or panic and build the confidence in designing the filter through this activity.
8. Communicate effectively by sharing their views among the teammates

Time Allotted for collaborative activity: 2 periods (50 minutes each)

Justification:

1 Period: (50 Minutes)

- ❖ The group name will be selected based on the Random name picker created using class tools website shown in the link

https://www.classtools.net/random-name-picker/81_c4YLM8

One students from each group will come and spin for selecting the group name once the name is selected it will be removed and seating arrangement will be made in the first **5 minutes**.

- ❖ Then in next **5 minutes**, TAI activity will be elaborated to the students with the learning objective and outcomes of the activity. The ground rules for the group interaction will be clearly told.
- ❖ Handouts with materials, Problem to be solved, ground rules and assessment rubrics will be distributed to the group in the next **3 minutes**.

Problem:

PART A:

A band reject FIR filter of length seven is required. It is to have lower and upper cut-off frequencies as 3 KHz and 6 KHz respectively. The sampling frequency is 18 KHz. Determine the filter co-efficients using Hanning window. Draw the structure of the filter.

The Part A is divided into four modules

Module 1: Find the desired impulse response

Module 2: Find the window sequence based on type of the problem

Module 3: Find the impulse response by multiplying desired impulse response & window sequence

Module 4: Find system transfer function and draw the structure of Filter using direct form and Linear phase structure.

PART B:

Simulate the FIR filter using Windowing technique using Matlab for LPF, HPF, BPF, BSF.

Ground rules for Group Interaction: (Reading - 2 minutes)

1. Each member of the group should first discuss the procedure for solving the problem.
2. The group members should be responsible in completing the task – don't waste time by simply chatting.
3. Solve the first module of the problem individually and verify the answers. The members of the group can proceed to next module once all the members of the group complete the module.
4. The doubts of group member should be clarified and make them to complete the task.
5. After completion of the task, the team members should prepare a chart with the Answers of each module, the part you felt to be difficult while designing the filter, Rate your team coordination and contribution by each member.
6. Any one member from the group should present in front of other members.

Time Allotment:

First Period (50 minutes)	
Team Name and seating arrangement	5 Minutes
Explaining TAI Activity	5 Minutes
Distribution of Handouts	3 Minutes
Reading the ground rules	2 Minutes
Opening Activity	5 Minutes
Solving Module 1 , 2 & 3	20 Minutes
Visual Quiz	6 Minutes
Conclusion – Discussing the answers	4 Minutes
Total	50 Minutes
Second Period (50 Minutes)	
Solving Module 4	8 Minutes
Simulating the FIR filter using Matlab	20 Minutes
Presentation by each group	20 Minutes
Concluding the activity by collecting the feedback from the individual members	2 Minutes
Total	50 Minutes
Conduct of online Quiz through course website after completion of activity	

Opening activity: (5 minutes)

Specification	Value	Unit
N- Filter Length		
F – Sampling Frequency		
f_l – lower cutoff frequency		
f_u – upper cutoff frequency		
ω_l – lower cutoff frequency		
ω_u – upper cutoff frequency		

**Department of Electronics and Communication Engineering****Academic Year 2021 – 2022 (Odd Semester)****Degree, Semester & Branch: V Semester B.E. ECE A****Course Code & Title: EC8553 & Discrete Time Signal Processing****Name of the Faculty member (s): Mrs.S.Jeeva****Innovative Practice Description**

- Unit / Topic: IV Finite Wordlength Effects – Fixed and Floating Point representation**

- Course Outcome:**

CO4 : The students will be able to Analyze the finite Word length effects in digital filters.

- Topic Learning Outcome:**

TLO20 : Represent in fixed point representation, floating point representation

- Activity Chosen: Jig Saw**

- Justification:**

The JigSaw activity is a method of organizing classroom activity that makes students dependent on each other to succeed. That enables each student of a "home" group to detail study of one topic. The topic contains number system, Binary, Decimal conversion. These topics are already learnt in III semester EC8392 Digital Electronics course. This activity is used to recall the basic concepts easily.

- Time Allotted for the Activity: 50 minutes**

- Details of the Implementation:**

At first the activity related contents/materials are shared to students through canvas (LMS). The activity topic is divided into number of subtopics. The same number of different shapes and members are chosen for group. The same number of different shape students are allotted with particular topic. The same shape of different number students are combined together to form Home group. They are discussed their own subtopic concepts in home group members. All the members in home group learnt about all the topics through expert group members. At last one member from each shape group summarize the points.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO3
CO4	2	3	3	2	3	3	2

- PO / PSO mapped:**

Innovative practice	PO1	PO9	PO10	PSO3
	2	3	3	2
Justification	Finite word A	Collaborative	The topics deals	The concept of

for correlation	length effects requires the knowledge of engineering fundamentals such as representation of numbers, Decimal to binary and binary to decimal conversion. This course outcome is moderately correlated.	activity – Jigsaw helps the students to Present results as a team, with smooth integration of contributions from all individual efforts. The course outcome is highly correlated.	with presentations of fixed- and floating-point representation. This task is accomplished through Jigsaw activity therefore course outcome is correlated highly.	finite word length effects such as quantitation noise, limit cycle oscillations are required for design, analyze VLSI and communication systems. So, the course outcome is moderately correlated.
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• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

Students are very interesting to work together and learn to ask questions to clarify their doubts. Most of the students enjoyed the discussion in home group and expert group.

❖ ***Benefit of the practice:***

This activity encourages collaborative learning among students, it will help the students communication & problem solving skills. This activity encourages groups to work harder to ensure that all members learn. The success of the activity was evaluated by asking the same model question in Internal Assessment test III – Around 80% of students answered correct.

❖ ***Challenges faced in implementation:***

Few students are hesitating to form the group with girls and vice versa. It takes additional 15 minutes to complete that activity compared to planned duration

References:

- <https://www.jigsaw.org/>
- https://www.newcastle.edu.au/data/assets/pdf_file/0016/109600/Jigsaw-learning-activity.pdf
- https://www.slideshare.net/PeriyanayagiS/finite-word-length-effects?qid=7a71a772-7b18-46bc-a2e7-6d6191058a97&v=&b=&from_search=1
- John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
- Tarun Kumar Rawat, 'Digital Signal Processing', Oxford. (Page No: 812-818)

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021 - 2022 (odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE A & B

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member: Dr.S.Periyanayagi & Mrs.S.Jeeva

Topic: Fixed Point and Floating Point Number Representation

Learning Outcomes:

The students will be able to

- ❖ Represent a number in fixed point and floating point representation.
- ❖ Differentiate fixed point and floating point representation.
- ❖ Participate actively in the learning process.
- ❖ Discuss the concepts, convey ideas, share the views to really understand the Fixed and floating point representation.
- ❖ Communicate effectively by sharing their views among the teammates.

Justification for choosing the topic for the activity:

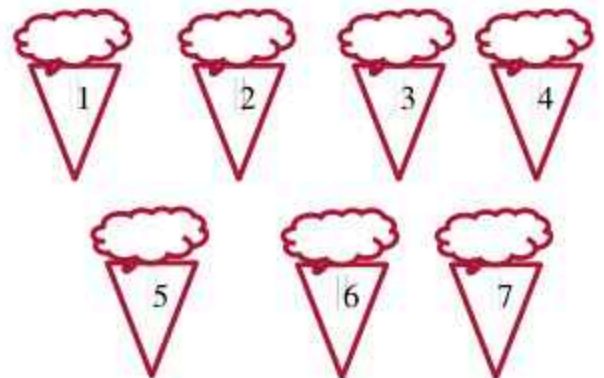
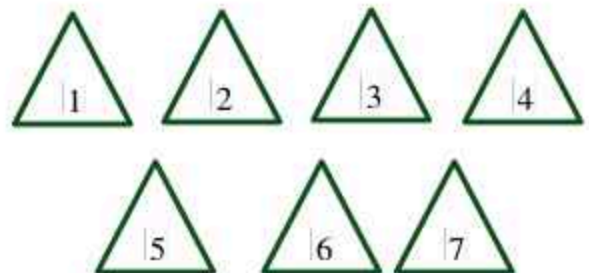
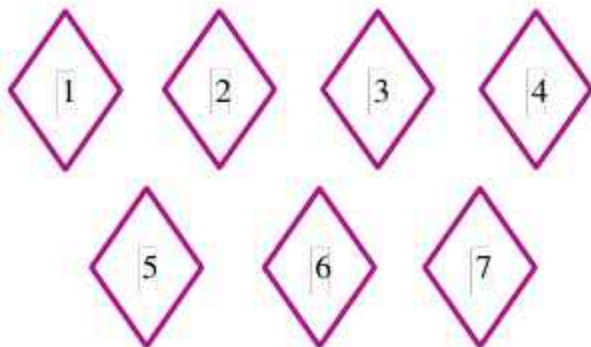
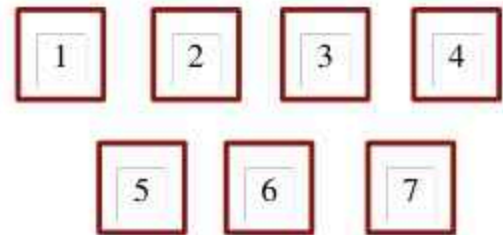
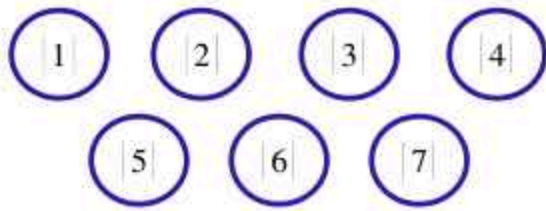
The concept of number system – Binary to decimal, Decimal to binary, one's complement and two's complement are already learnt in Digital electronics course. This topic is related to the same concepts and easy to understand. The activity '**JIGSAW**' is used to recall the concepts learnt and enhance the learning through this collaboration.

Plan for Implementation:

- ❖ The material related to the topic, video will be posted 1 week before the conduct of activity.
- ❖ **Activity: JIGSAW-** The Class Strength is divided into 7 Shapes Groups with 7 members as shown below using Random Lot System.
- ❖ Each group is allotted with a shape and its known as Home group as shown below.
- ❖ 6 Shapes are named 1,2,3,4,5,6,7. All the 1's in different Shapes are made to form Expert group.
- ❖ Each Expert group is allotted with one topic
 - ✓ Number System – Binary and Decimal
 - ✓ Addition and Multiplication in Decimal and Binary Number system
 - ✓ Fixed Point Representation Sign magnitude, One's Complement
 - ✓ Fixed Point Representation Two's Complement
 - ✓ Floating Point Representation
 - ✓ Fixed point representation Addition and Multiplication.
- ❖ The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas.
- ❖ The Expert group members should go to their original shape group and discuss the points (30 Minutes) to the other members.
- ❖ All the members in home group learnt about all the topics through expert group members.
- ❖ Finally one from each shape group should summarize the points of the topic Fixed and Floating point representation to the class (20 Minutes).
- ❖ Review of points and conclusion of the activity (5 minutes)

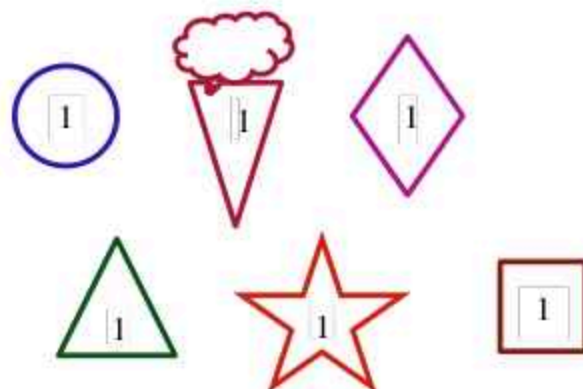
- ❖ Thus the concept of fixed point and floating point representation is known to all by involving the students actively.

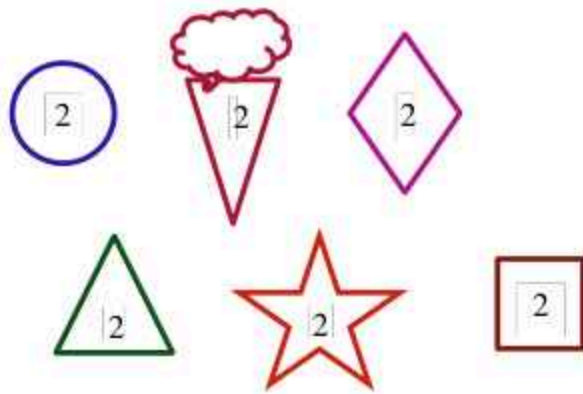
Home Group formation:



- ❖ Expert group consist of members from each shape and assigned with a topic for discussion (10 Minutes) which has been already learnt in III Semester subject.

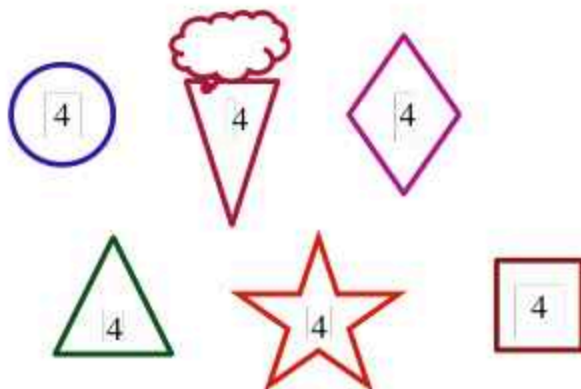
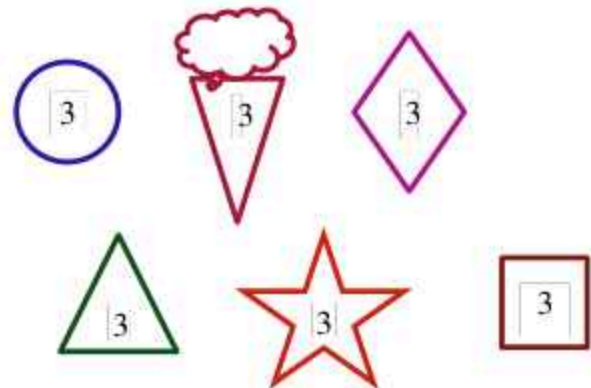
GROUP 1
Topic: Number System – Binary
 and Decimal



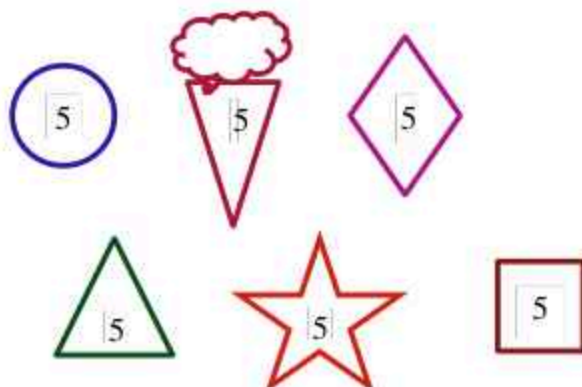


GROUP 2
Topic: Addition and
Multiplication in Decimal and
Binary Number system

GROUP 3
Topic: Fixed Point representation
Sign magnitude, One's
Complement

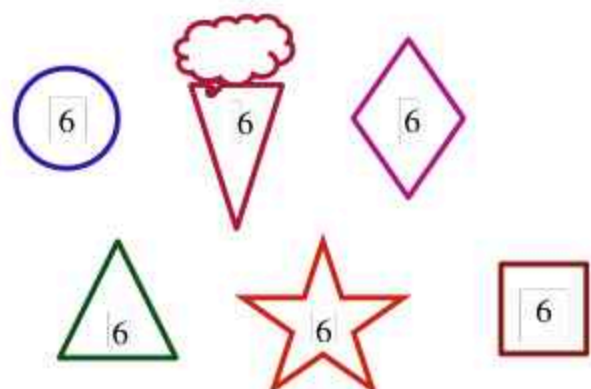


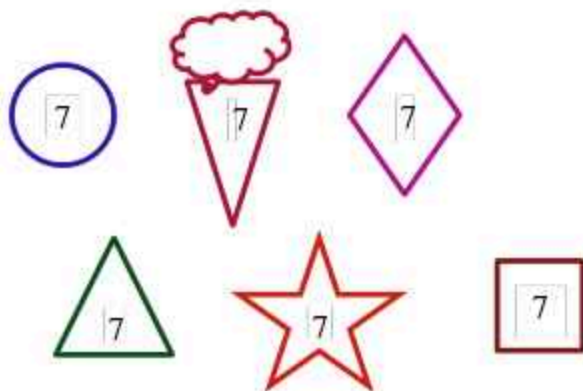
GROUP 4
Topic: Fixed Point representation
Two's Complement



GROUP 5
Topic: Floating Point
representation

GROUP 6
Topic: Fixed point representation
Addition and Multiplication.





Summarize the points of Fixed Point & Floating Point representation

References:

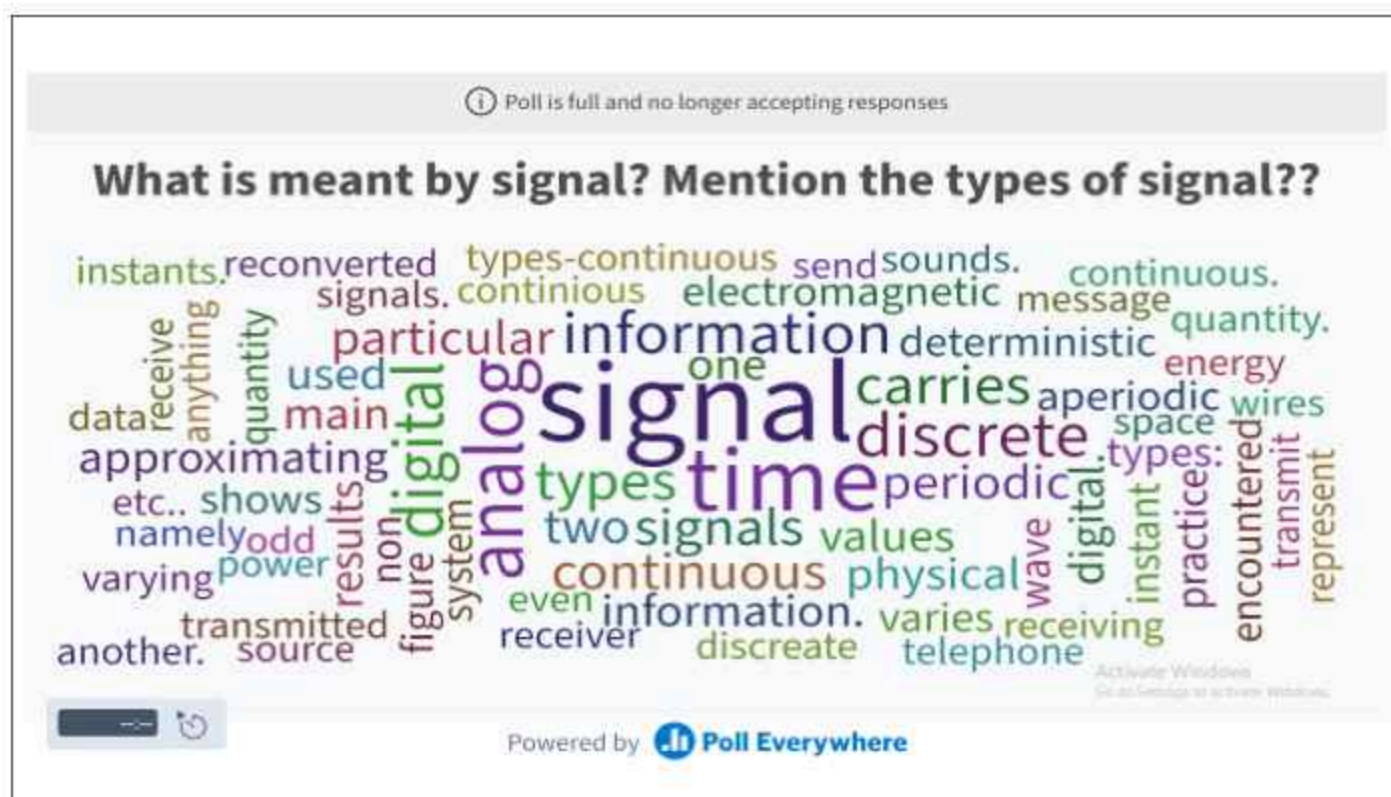
Kindly refer the following materials before coming to the Class

- <https://www.jigsaw.org/>
- https://www.newcastle.edu.au/data/assets/pdf_file/0016/109600/Jigsaw-learning-activity.pdf
- https://www.slideshare.net/PeriyanayagiS/finite-word-length-effects?qid=7a71a772-7b18-46bc-a2e7-6d6191058a97&v=&b=&from_search=1
- John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
- Tarun Kumar Rawat, 'Digital Signal Processing', Oxford. (Page No: 812-818)

INSTRUCTIONAL METHODS

Name of the Faculty member: Mrs.S.Jeeva

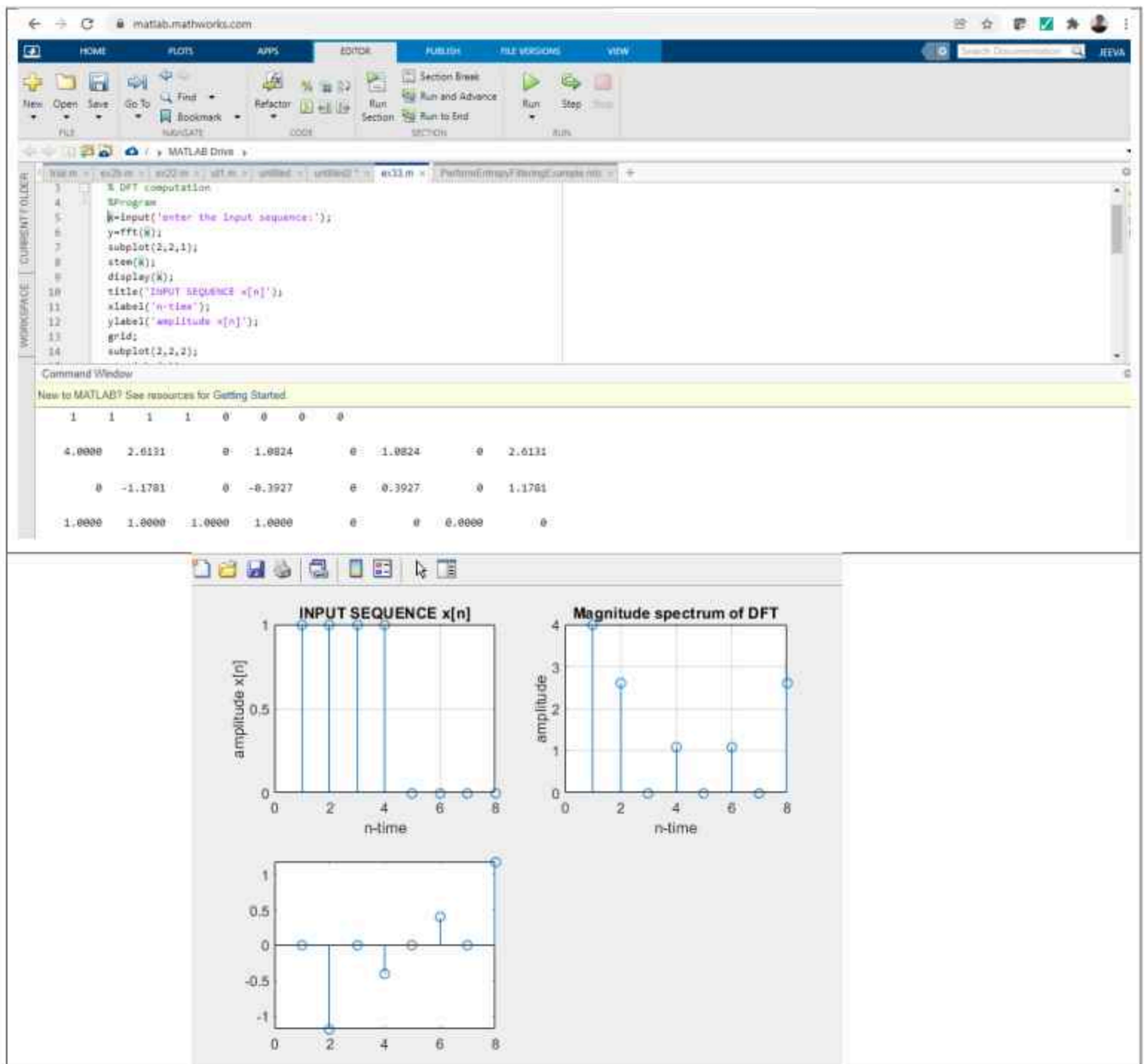
Date: 23.08.2021



Name of Instructional Method: Online Simulation -MATLAB

Topic: 8 Point DIT FFT

Date: 04.09.2021



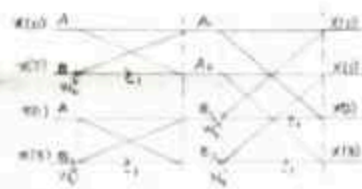
Name of Instructional Method: One Minute Paper

Topic: Linear Filtering using FFT

Date: 08.09.2021

Linear Filtering using FFT

Step 1: Find $X(k)$ using DIT-FFT



Step 2: Find $H(k)$ using DIT-FFT



Step 3: $Y(k) = X(k) \cdot H(k)$

Step 4: Find $y(n)$ using IDFT using DIT-FFT



$$y(n) = \frac{1}{N} \{y(n), y(n), y(n), y(n)\}$$

Reg.no: 953619106016

Name: A.Elavarasi

Class: ECE-A

Linear Filtering using FFT

Step 1: Find $X(k)$ using DIT-FFT



Step 2: Find $H(k)$ using DIT-FFT



Step 3: $Y(k) = X(k) \cdot H(k)$

Step 4: Find $y(n)$ using IDFT using DIT-FFT



$$y(n) = \frac{1}{N} \{y(n), y(n), y(n), y(n)\}$$

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021- 2022 (Odd Semester)

INSTRUCTIONAL METHODS

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member: Mrs.S.Jeeva

UNIT II INFINITE IMPULSE RESPONSE FILTER

Name of Instructional Method: Think Pair Share

Topic: Characteristics of commonly used analog filters

Date: 13.09.2021

Think Pair Share

13.09.2021



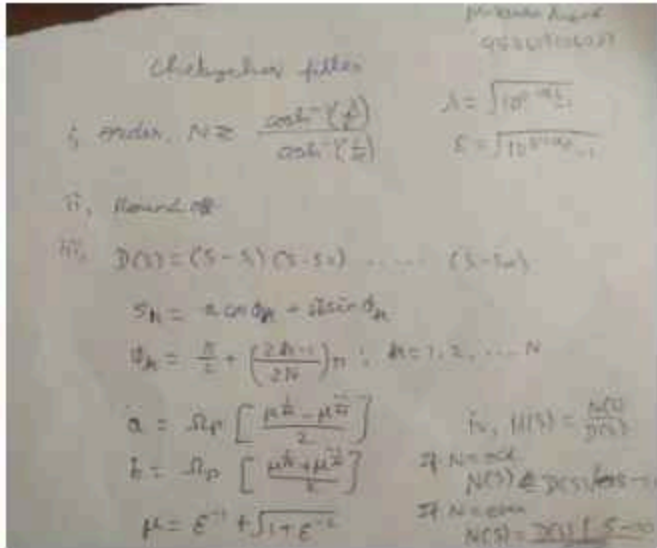
Characteristics of Analog & Digital
Filters



Name of Instructional Method: Exit Slips

Topic: Analog Chebyshev filter Design

Date: 20.09.2021



Chebyshev filter

i. order, $N \geq \frac{\cosh^{-1}(\frac{1}{\epsilon})}{\cosh^{-1}(\frac{1}{\epsilon})}$

ii. Bandwidth

iii. $D(s) = (s-s_1)(s-s_2) \dots (s-s_N)$

$S_k = a \cos \theta_k + j b \sin \theta_k$

$\theta_k = \frac{\pi}{N} + \left(\frac{2k-1}{2N}\right)\pi, k=1, 2, \dots, N$

$a = \Omega_p \left[\frac{\mu^{\frac{1}{N}} - \mu^{\frac{N-1}{N}}}{2} \right]$

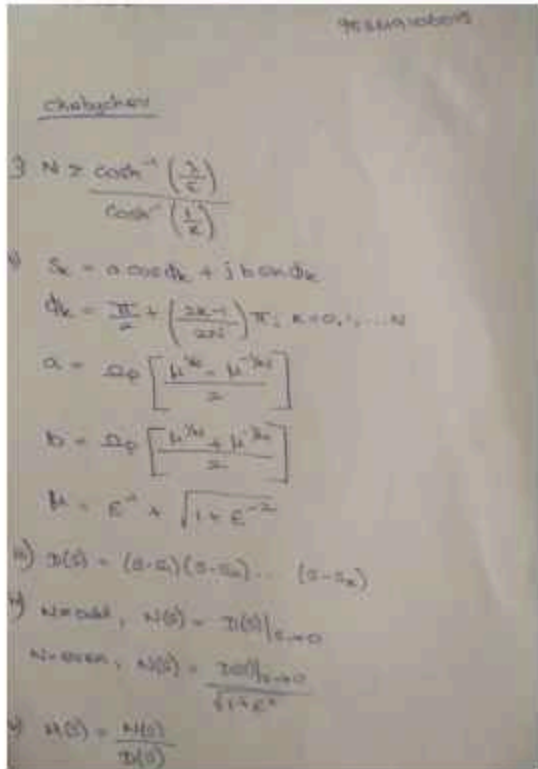
$b = \Omega_p \left[\frac{\mu^{\frac{1}{N}} + \mu^{\frac{N-1}{N}}}{2} \right]$

$\mu = \epsilon^{-1} + \sqrt{1 + \epsilon^{-2}}$

iv. $H(s) = \frac{N(s)}{D(s)}$

If N is odd, $N(s) \in \mathbb{R}$ (passive)

If N is even, $N(s) = \sqrt{1 + \epsilon^{-2}}$



Chebyshev

3. $N \geq \cosh^{-1} \left(\frac{1}{\epsilon} \right)$

$\cosh^{-1} \left(\frac{1}{\epsilon} \right)$

4. $S_k = a \cos \theta_k + j b \sin \theta_k$

$\theta_k = \frac{\pi}{N} + \left(\frac{2k-1}{2N}\right)\pi, k=0, 1, \dots, N$

$a = \Omega_p \left[\frac{\mu^{\frac{1}{N}} - \mu^{\frac{N-1}{N}}}{2} \right]$

$b = \Omega_p \left[\frac{\mu^{\frac{1}{N}} + \mu^{\frac{N-1}{N}}}{2} \right]$

$\mu = \epsilon^{-1} + \sqrt{1 + \epsilon^{-2}}$

5. $D(s) = (s-s_1)(s-s_2) \dots (s-s_N)$

6. If N is odd, $N(s) = D(s)|_{s \rightarrow 0}$

If N is even, $N(s) = \frac{D(s)|_{s \rightarrow 0}}{\sqrt{1 + \epsilon^{-2}}}$

7. $H(s) = \frac{N(s)}{D(s)}$

Name of Instructional Method: Four corner Four Question

Topic: Design of IIR filter from analog filter – Bilinear Transformation

Date: 04.10.2021

FOUR CORNER FOUR QUESTION

Pre-warping Formula

Relationship between S-plane and Z-Plane mapping using Bilinear Transformation

Design Procedure of Digital filter using bilinear transformation

Advantages of Bilinear transformation

1. Pre-warping formula

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p T}{2}\right)$$
$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s T}{2}\right)$$

2. Relation

$$S = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

3. Advantages

Bilinear transformation has no aliasing problem one-to-one mapping

$$H_d(z) = H(s) \Big|_{s=\frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]}$$

1. Pre-warping formula

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p T}{2}\right)$$
$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s T}{2}\right)$$
$$S = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

* No aliasing
* stable CT system can be mapped in stable digital system

$$H_d(z) = H(s) \Big|_{s=\frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]}$$

Reg No: 9C3619106029

Name of Instructional Method: Visible Quiz

Topic: Revision

Date: 08.10.2021



RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021- 2022 (Odd Semester)

INSTRUCTIONAL METHODS

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

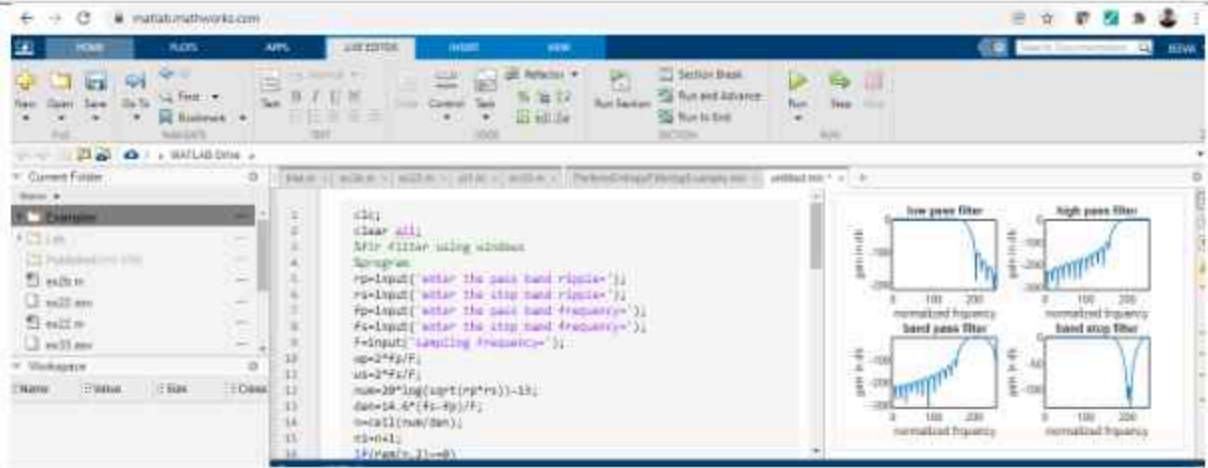
Name of the Faculty member: Mrs.S.Jeeva

UNIT III FINITE IMPULSE RESPONSE FILTER

Name of Instructional Method: Simulation Tool

Topic: Filter Design Using Windowing techniques – Hanning window

Date: 22.10.2021

	
<pre> B=fir1(n,wp,y); [h,a]=freqz(B,1,256); M=20*log(abs(h)); subplot(2,2,1); plot(M); title('low pass filter'); xlabel('normalized frequency'); ylabel('gain in db'); B=fir1(n,wp,'high',y); [h,a]=freqz(B,1,256); M=20*log(abs(h)); subplot(2,2,2); plot(M); title('high pass filter'); xlabel('normalized frequency'); ylabel('gain in db'); wn=[wp,ws]; B=fir1(n,wn,y); [h,a]=freqz(B,1,256); M=20*log(abs(h)); subplot(2,2,3); plot(M); </pre>	<pre> enter the pass band ripple= 18 enter the stop band ripple= 29 enter the pass band frequency= 0.7 enter the stop band frequency= 0.9 sampling frequency= 2 </pre>

Name of Instructional Method: Concept Check

Topic: Frequency Sampling Techniques

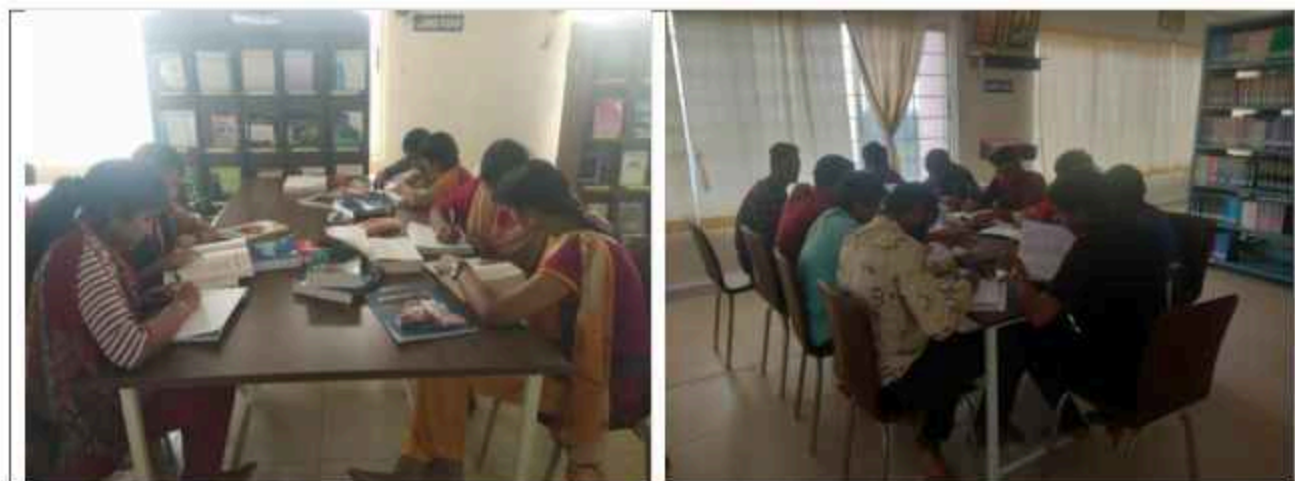
Date: 27.10.2021

RAMCO INSTITUTE OF TECHNOLOGY Department of Electronics and Communication Engineering Academic Year: 2021 - 2022 (2021 Semester) Degree, Semester & Branch: V Semester E.E.E & E.E.E Course Code & Title: ECE505 Advanced Time Signal Processing Name of the Faculty member: Dr.S.Prasanna & Mrs.S.Joshi Topic: FIR Filter Design Activity: Concept Note 1. The Desired Impulse response of the FIR Filter $h_d(n) = \int_{-\infty}^{\infty} H_d(e^{j\omega}) e^{j\omega n} d\omega \quad \text{---} \quad h_d(n) = \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} \frac{d\omega}{2\pi}$ 2. The transfer function $H(z) = \sum_{n=0}^{N-1} h_d(n) (z^{-1})^n$ 3. Realizable transfer function $H(z) = z^{-(N-1)/2} \sum_{n=0}^{N-1} h_d(n) z^{-n} \quad \text{---} \quad H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=0}^{N-1} h_d(n) e^{-j\omega n}$ 4. Give the correct Window sequence Hamming Window: $W_H(n) = \begin{cases} 0.54 + 0.46 \cos\left(\frac{2\pi n}{N-1}\right) & 0 \leq n \leq \frac{N-1}{2} \\ 0 & \text{otherwise} \end{cases} \quad W_H(e^{j\omega}) = \frac{1}{2} \left[\frac{\sin\left(\frac{\omega(N-1)/2 + \omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} + \frac{\sin\left(\frac{\omega(N-1)/2 - \omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} \right]$ Rectangular Window: $W_R(n) = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases} \quad W_R(e^{j\omega}) = \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} e^{-j\omega(N-1)/2}$ 5. Frequency sampling method $h(n) = \frac{1}{N} \left[H(e^{j\omega}) \sum_{k=0}^{N-1} e^{-j\omega_k n} \right] \quad \text{---} \quad h(n) = \frac{1}{N} \sum_{k=0}^{N-1} H(e^{j\omega_k}) e^{j\omega_k n}$ Name of the student: <u>N. Shilpa</u> Register Number: <u>2021EE000000000000</u> Signature of the student: <u>N. Shilpa</u>	RAMCO INSTITUTE OF TECHNOLOGY Department of Electronics and Communication Engineering Academic Year: 2021 - 2022 (2021 Semester) Degree, Semester & Branch: V Semester E.E.E & E.E.E Course Code & Title: ECE505 Advanced Time Signal Processing Name of the Faculty member: Dr.S.Prasanna & Mrs.S.Joshi Topic: FIR Filter Design Activity: Concept Note 1. The Desired Impulse response of the FIR Filter $h_d(n) = \int_{-\infty}^{\infty} H_d(e^{j\omega}) e^{j\omega n} d\omega \quad \text{---} \quad h_d(n) = \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} \frac{d\omega}{2\pi}$ 2. The transfer function $H(z) = \sum_{n=0}^{N-1} h_d(n) (z^{-1})^n$ 3. Realizable transfer function $H(z) = z^{-(N-1)/2} \sum_{n=0}^{N-1} h_d(n) z^{-n} \quad \text{---} \quad H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=0}^{N-1} h_d(n) e^{-j\omega n}$ 4. Give the correct Window sequence Hamming Window: $W_H(n) = \begin{cases} 0.54 + 0.46 \cos\left(\frac{2\pi n}{N-1}\right) & 0 \leq n \leq \frac{N-1}{2} \\ 0 & \text{otherwise} \end{cases} \quad W_H(e^{j\omega}) = \frac{1}{2} \left[\frac{\sin\left(\frac{\omega(N-1)/2 + \omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} + \frac{\sin\left(\frac{\omega(N-1)/2 - \omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} \right]$ Rectangular Window: $W_R(n) = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases} \quad W_R(e^{j\omega}) = \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} e^{-j\omega(N-1)/2}$ 5. Frequency sampling method $h(n) = \frac{1}{N} \left[H(e^{j\omega}) \sum_{k=0}^{N-1} e^{-j\omega_k n} \right] \quad \text{---} \quad h(n) = \frac{1}{N} \sum_{k=0}^{N-1} H(e^{j\omega_k}) e^{j\omega_k n}$ Name of the student: <u>N. Shilpa</u> Register Number: <u>2021EE000000000000</u> Signature of the student: <u>N. Shilpa</u>
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Name of Instructional Method: Open Book Test

Topic: Revision

Date: 01.11.2021



RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021- 2022 (Odd Semester)

INSTRUCTIONAL METHODS

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member: Mrs.S.Jeeva

UNIT IV FINITE WORDLENGTH EFFECTS

Name of Instructional Method: Concept Test

Topic: Truncation & Rounding Errors

Date: 12.11.2021

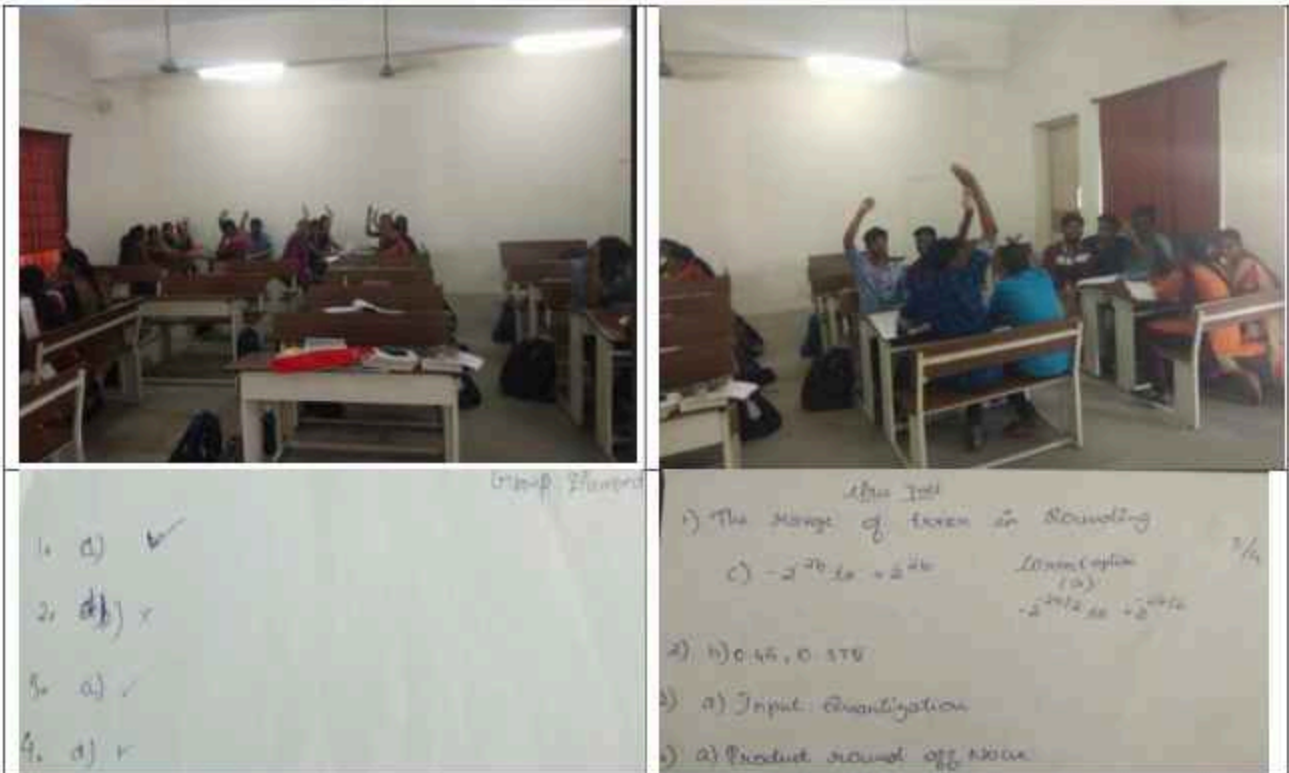
Truncation Error Range		
Type of Quantization	Number representation	Range of numbers
Truncation	Positive numbers and its complementaries	$-2^{n-1} \leq 0$
Truncation	Signed magnitudes and its complementaries	$0 \leq 2^{n-1} \leq 2^n$
Rounding	All positive and negative numbers	$-2^{n-1} \leq 2^{n-1}$
Floating point numbers		
Truncation	Positive numbers and signed magnitudes and its complementaries	$-2^{n-1} \leq 0$
Truncation	Two's complement negative	$0 \leq 2^{n-1} \leq 2^n$

Truncation Error		
Effect of Rounding		
$-\frac{1}{2} \Delta \leq e \leq \frac{1}{2} \Delta$		
Fixed Point	Positive numbers & its complementaries	$-2^{n-1} \leq 0$
Fixed Point	Signed magnitudes & its complementaries	$0 \leq 2^{n-1} \leq 2^n$
Rounding	All positive and negative numbers	$-2^{n-1} \leq 2^{n-1}$
Floating point		
Truncation	(+ve & its complementaries)	$-2^{n-1} \leq 0$
Truncation	(-ve & its complementaries)	$0 \leq 2^{n-1} \leq 2^n$
Rounding	All (+ve & (-ve) numbers	$-2^{n-1} \leq 2^{n-1}$

Name of Instructional Method: Class Poll

Topic: Coefficient Quantization Error

Date: 17.11.2021

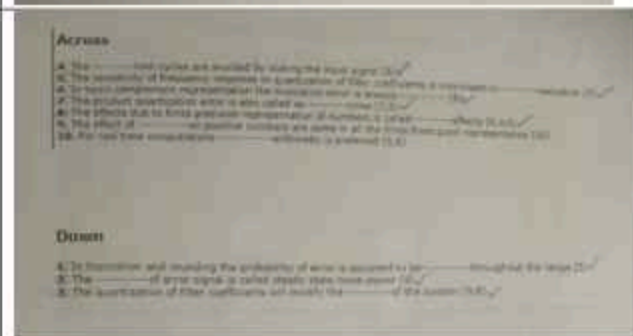
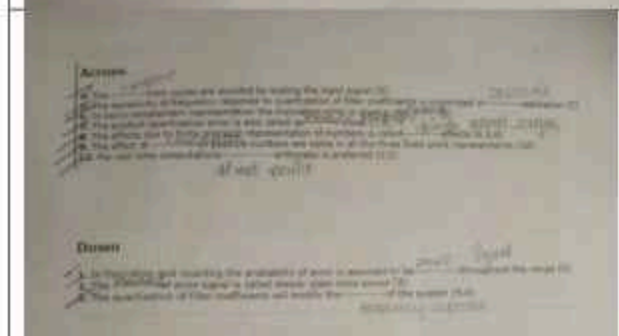


Name of Instructional Method: Crossword Puzzle

Topic: Scaling to prevent Overflow & Revision

Date: 19.11.2021





RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021- 2022 (Odd Semester)

INSTRUCTIONAL METHODS

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

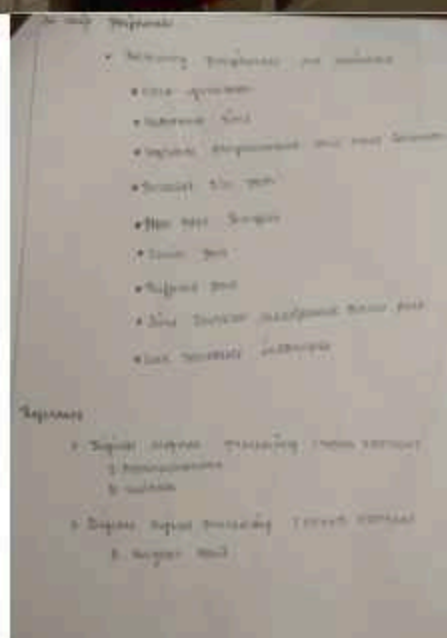
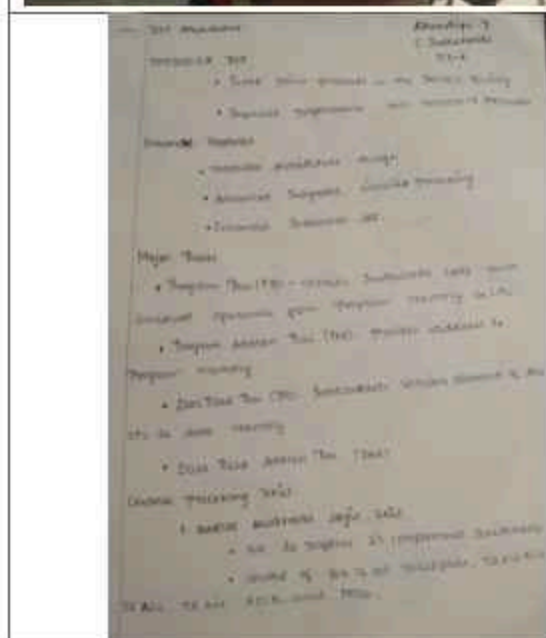
Name of the Faculty member: Mrs.S.Jeeva

UNIT V INTRODUCTION TO DSP PROCESSORS

Name of Instructional Method: Reflective Journal

Topic: DSP Architecture

Date: 26.11.2021



Name of Instructional Method: Flipped Class

Topic: Application Examples

Date: 30.11.2021

